MiniRoute's Verification

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About the DUT

- MiniRoute is a simplified version of a network switch/router that routes packets between devices based on source and destination addresses.
- It supports configuration packets and different data packet such as broadcast, multicast, and one-to-one packets.

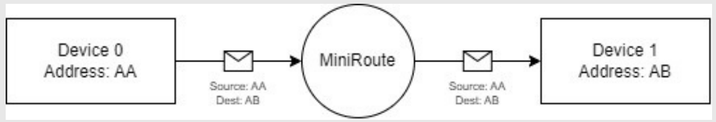

Fig.1. One-to-one packet

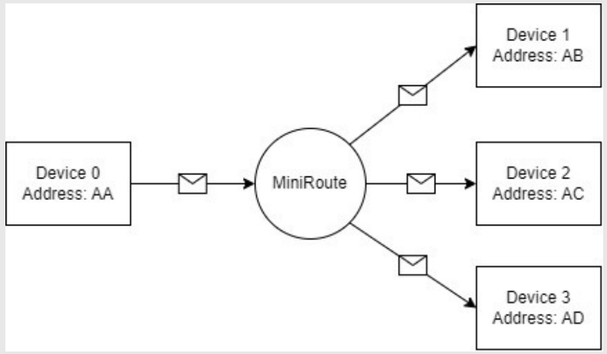

Fig.2. Multicast packet

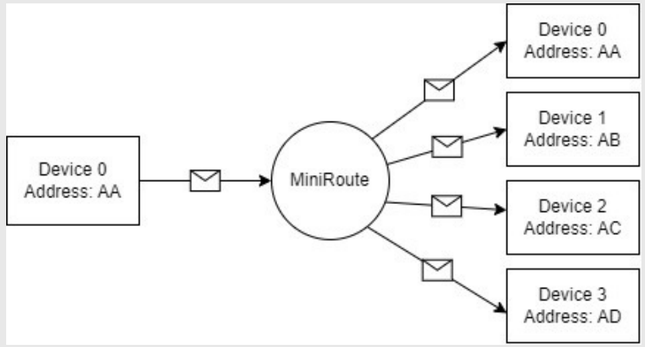

Fig.3. Broadcast packet

About the DUT

- A device can change its configuration through the following mechanism.

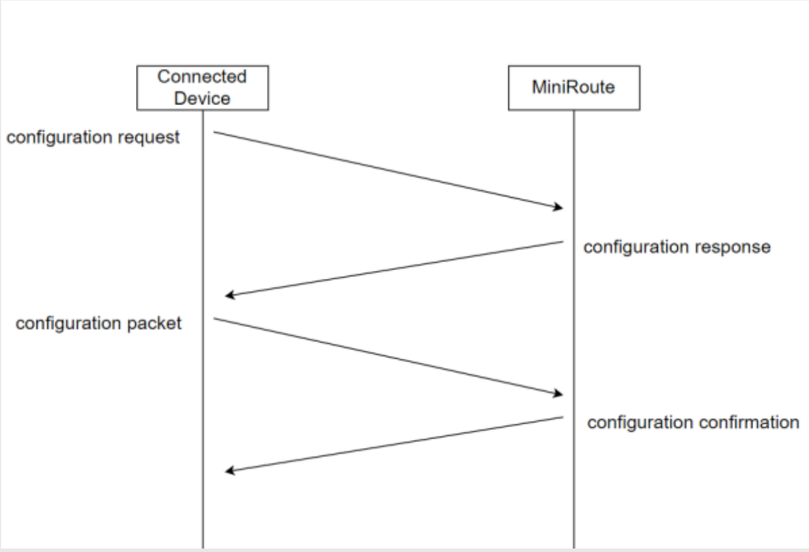

Fig.4. Successful Configuration Flow

About the DUT

- A packet consists of a header and payload.
- Header contains source and destination addresses, checksum, and packet type.
- Payload contains the data being transmitted.

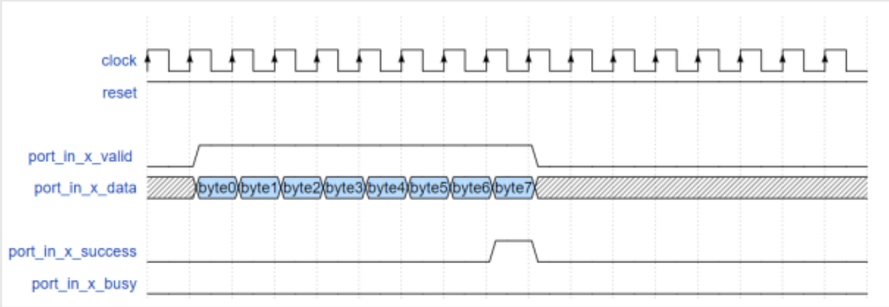

Fig.5. Input Packet

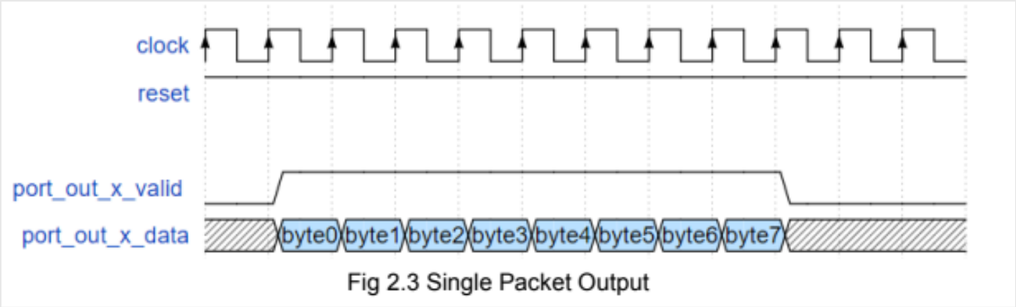

Fig.6. Output Packet

Verification Architecture

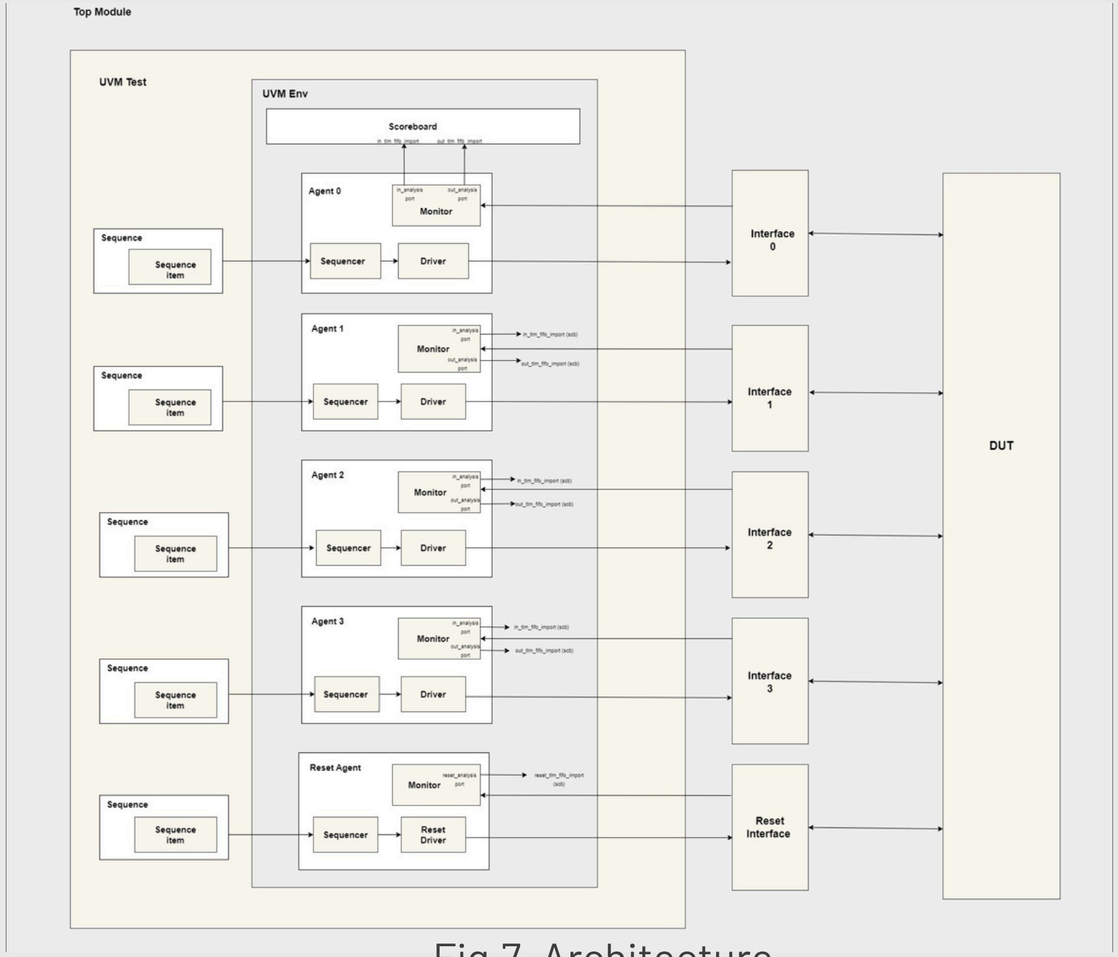

Fig.7. Architecture

Verification Plan

Verification Plan For MiniRoute											
Test #	Test Name	Test Type	Purpose	Stimulus	Passing Criteria	Port #	Date of completion	Total Monitored	Score = Fail / Total	Spec Reference	Comments
1	Sanity Trafic Test	Directed	To verify the basic funtionality of router. To check whether valid packets are processed correctly by the router.	Valid configuration request, configuration packets and random type data packets.	Packets should be routed to the correct destination port as specified by the packet type and configuration.	0,1,2,3	30/08/2024	80000	49/80000	3.b	All Test Passed except few cases
2	Configuration Test	Random	To validate the configuration request and response mechanism processes correctly.	Valid configuration request, configuration packets and random type data packets.	Packets should be routed to the correct destination port as specified by the packet type and configuration.	0,1,2,3	31/08/2024	80000	17/80000	3.b	All Test Passed except few cases
3	Random Packet Test	Random	To validate that dut doesn't process incorrect/ random configuration and data packets.	Invalid/random packets	Error packets should be generated with the correct format and content in response to invalid packets.	0,1,2,3	01/09/2024	80,000	103/80000	3.b	All Test Passed except few cases
4	Random Reset Test	Random	To check the behaviour of dut in case of random reset.	Valid packets with random reset	Configurations should be default after reset assertion. Output packets should receive only in the first 10 cycles of reset.	0,1,2,3	01/09/2024	27000	6/27000	1.c & 4	All test passed

Fig.8. Test Plan

Bugs Reported

- Priority Issue

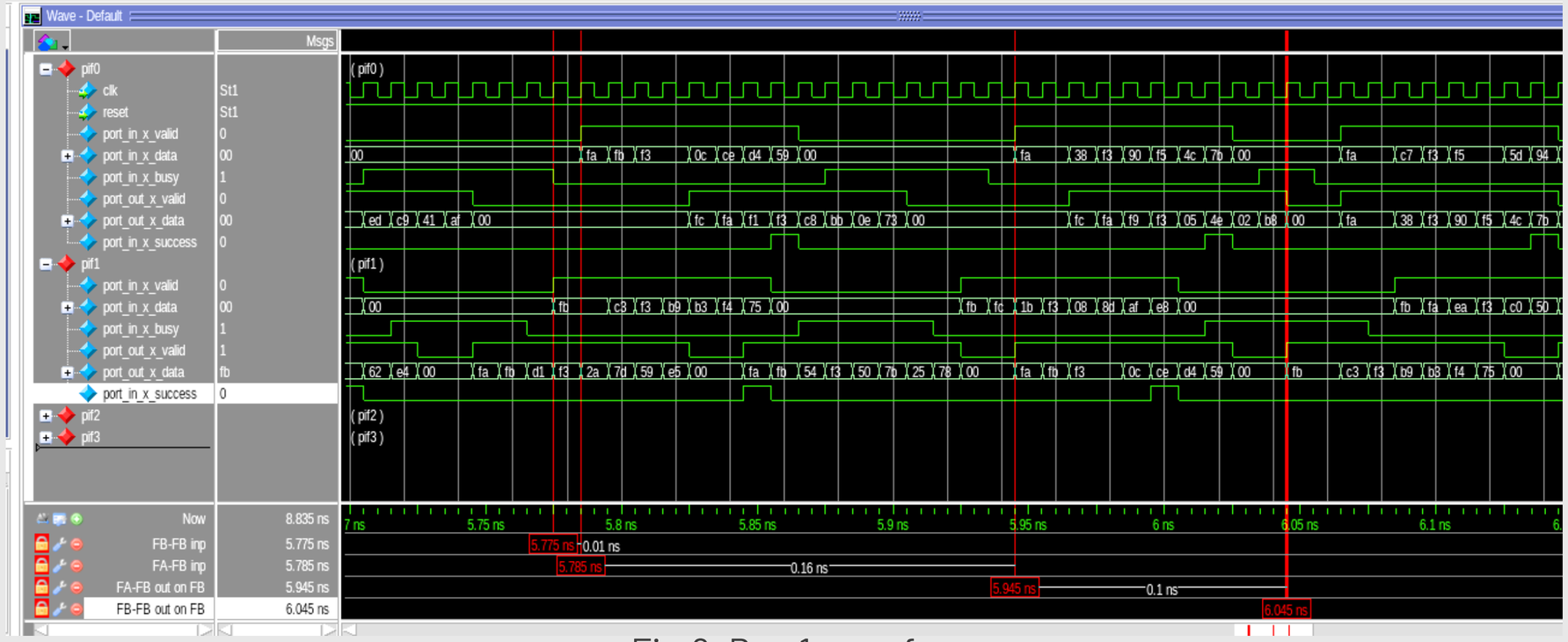

Fig.9. Bug 1 waveform

Bugs Reported

- port_x_success gets high once cycle late.
- Confirmation Packet Issue

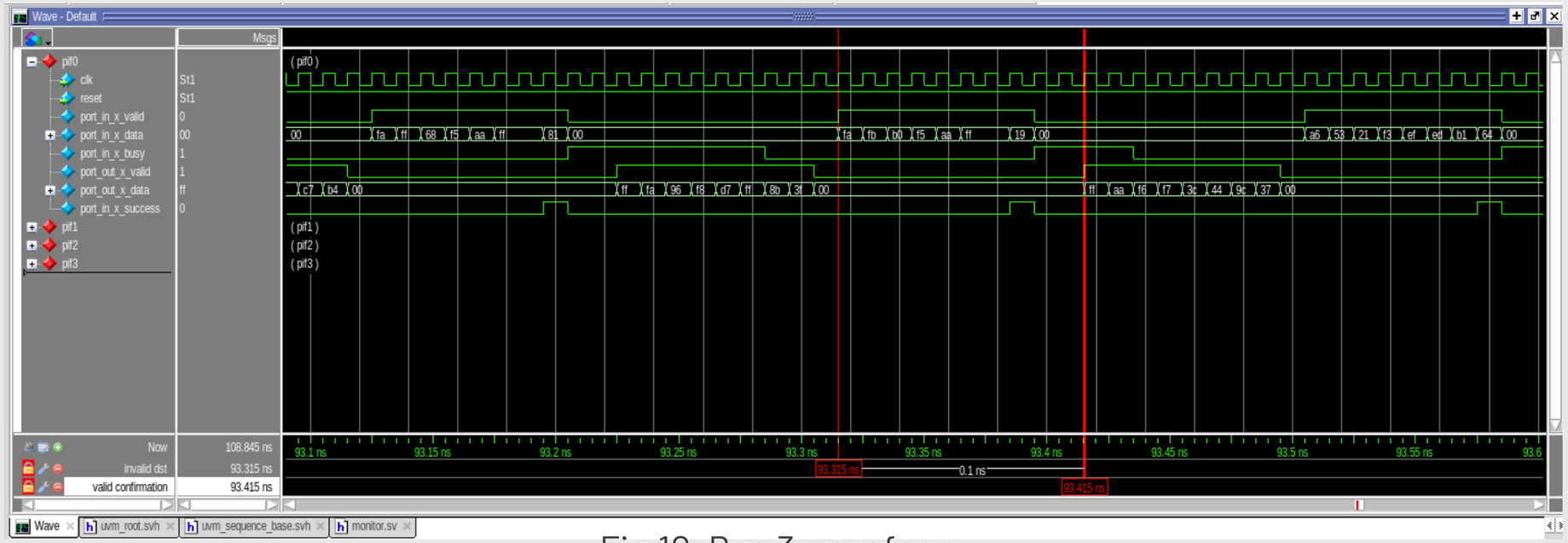

Fig.10. Bug 3 waveform

Additional Bugs

- Packet routes to the default address even after the configuration is changed.

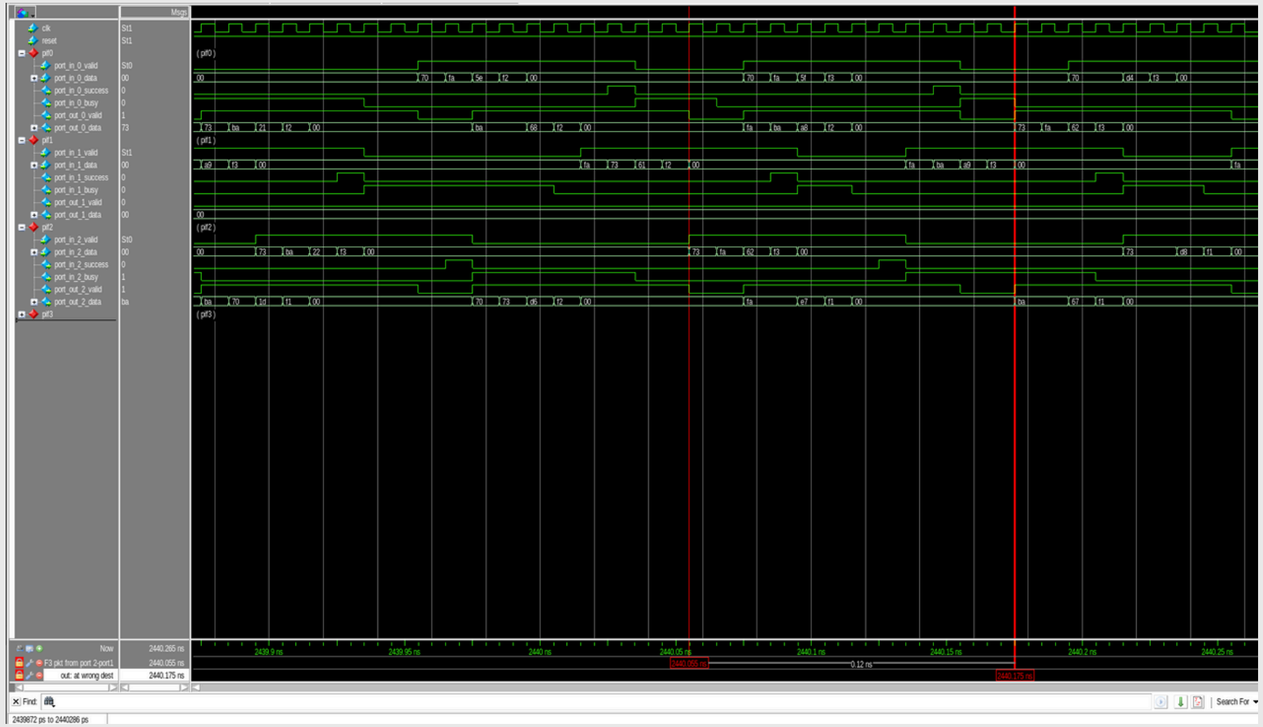

Fig.11. Bug waveform

Additional Bugs

- Packet routes to the port which is disconnected.

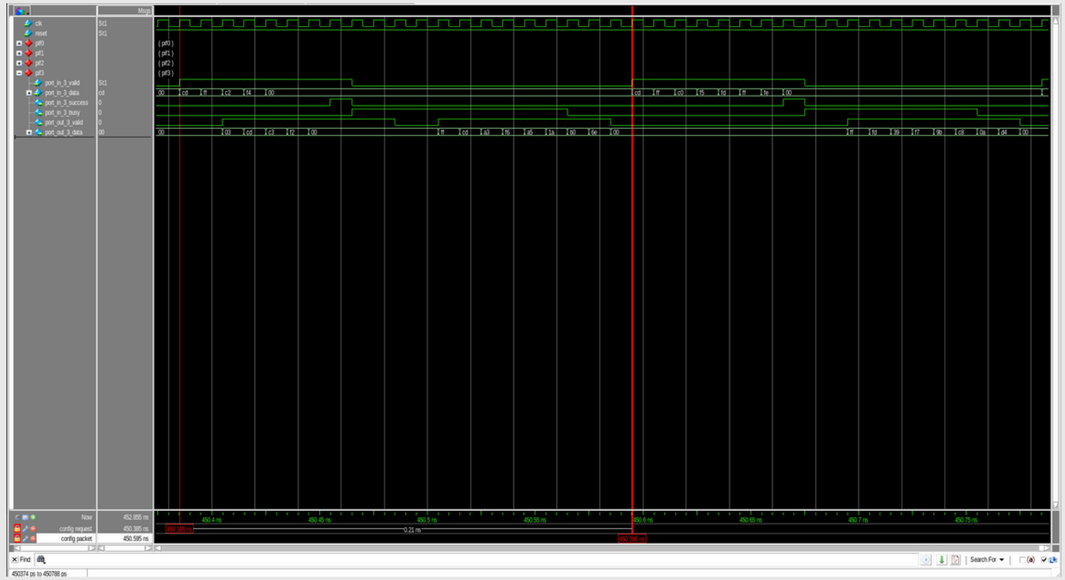

Fig.12.1. Bug waveform

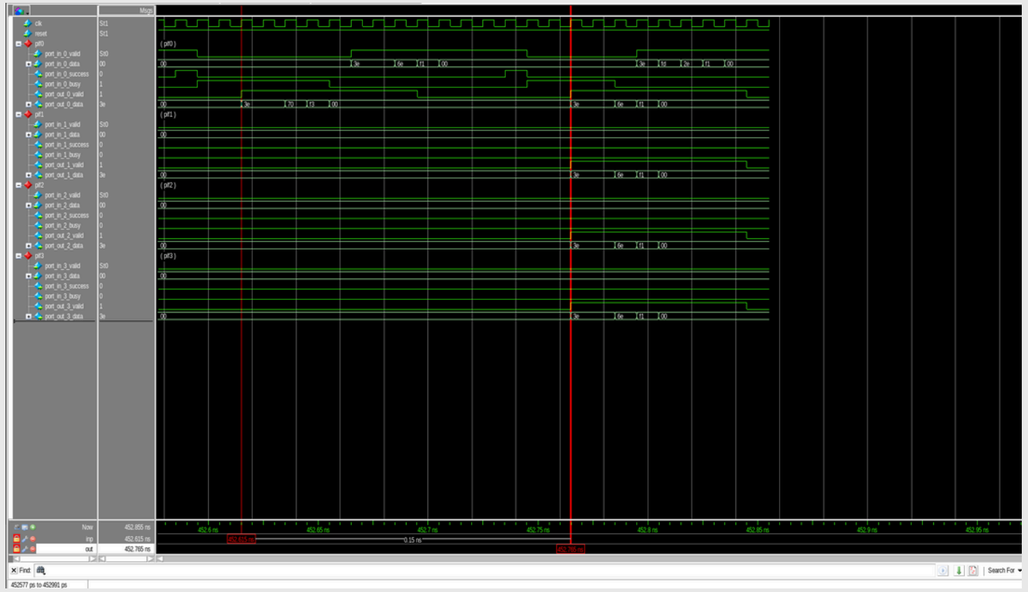

Fig.12.2. Bug waveform

Additional Bugs

- Packet routes to the port which is disconnected.

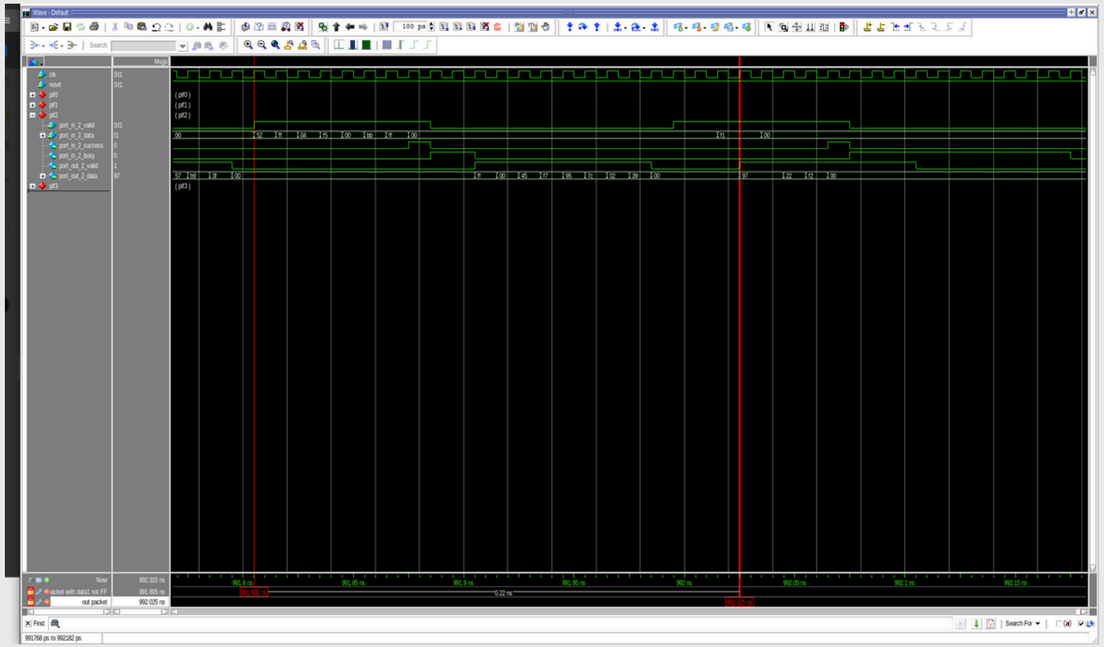

Fig.13. Bug waveform

Overall Results

- DUT performs well in routing and error detection under normal conditions but requires some improvements in handling corner cases during configuration and data transmission as well .
- It successfully routed almost 99% of one-to-one, broadcast, multicast and configuration packets.

Thank you!

Any Questions?